

HOMework 3

LINEAR ALGEBRA *

10-606 MATHEMATICAL FOUNDATIONS FOR MACHINE LEARNING

START HERE: Instructions

- **Collaboration Policy:** Please read the collaboration policy in the syllabus.
- **Late Submission Policy:** See the late submission policy in the syllabus.
- **Submitting your work:** You will use Gradescope to submit answers to all questions.
 - **Written:** For written problems such as short answer, multiple choice, derivations, proofs, or plots, please use the provided template. Submissions can be handwritten onto the template, but should be labeled and clearly legible. If your writing is not legible, you will not be awarded marks. Alternatively, submissions can be written in $\text{\LaTeX}$. Each derivation/proof should be completed in the boxes provided. To receive full credit, you are responsible for ensuring that your submission contains exactly the same number of pages and the same alignment as our PDF template.
 - **Latex Template:** <https://www.overleaf.com/read/ykdbpdvcczqr#a6dc7b>

Question	Points
Linear Systems and Linear Algebra	13
Matrix Memories	5
Eigenvectors and eigenvalues	7
Positive semidefinite matrices	8
Total:	33

*Compiled on Friday 11th September, 2026 at 17:42

Instructions for Specific Problem Types

For “Select One” questions, please fill in the appropriate bubble completely:

Select One: Who taught this course?

- ☒ Bryan Wilder
☐ Marie Curie
☐ Noam Chomsky

If you need to change your answer, you may cross out the previous answer and bubble in the new answer:

Select One: Who taught this course?

- ☒ Bryan Wilder
☐ Marie Curie
☒ Noam Chomsky

For “Select all that apply” questions, please fill in all appropriate squares completely:

Select all that apply: Which are scientists?

- ☒ Stephen Hawking
☒ Albert Einstein
☒ Isaac Newton
☐ I don't know

Again, if you need to change your answer, you may cross out the previous answer(s) and bubble in the new answer(s):

Select all that apply: Which are scientists?

- ☒ Stephen Hawking
☒ Albert Einstein
☒ Isaac Newton
☐ I don't know

For questions where you must fill in a blank, please make sure your final answer is fully included in the given space. You may cross out answers or parts of answers, but the final answer must still be within the given space.

Fill in the blank: What is the course number?

10-606

10-606~~7~~

1 Linear Systems and Linear Algebra (13 points)

1. For each statement, indicate whether it is true or false.
 - (a) (1 point) A system of n linear equations with n unknowns has at least one solution.
☐ True ☐ False
 - (b) (1 point) A system of n linear equations with n unknowns has at most one solution.
☐ True ☐ False
 - (c) (2 points) Rank of matrix product $\mathbf{AB}$ can be greater than the rank of either $\mathbf{A}$ or $\mathbf{B}$.
☐ True ☐ False
2. (3 points) Consider the following matrix:

$$\mathbf{B} = \begin{bmatrix} 1 & 3 & 1 & 3 \\ 4 & 4 & -1 & 3 \\ 0 & 0 & -2 & -1 \\ -2 & -1 & 0 & 0 \end{bmatrix}$$

Compute the matrix rank of **B**. You may do it by hand or by using Python and numpy. You must show your work. If you use Python and numpy, Just paste your working code in the Work box.

Rank

Work

3. (6 points) Rewrite the following system in matrix form and solve it by Gaussian Elimination.

$$\begin{array}{rcl} 3x_1 - 7x_2 - 2x_3 & = & 5 \\ -3x_1 + 5x_2 + x_3 & = & 0 \\ 6x_1 - 4x_2 & = & 1 \end{array}$$

Solution**Work**

2 Matrix Memories (5 points)

Matrix Memories store a single pattern pair $\mathbf{s} \rightarrow \mathbf{t}$ by encoding the outer product of a target vector $\mathbf{t}$ with the input stimulus $\mathbf{s}$. In this problem, we will examine exactly when it is possible to store multiple pattern pairs. Suppose we wish to store the K pattern pairs below. Note that the k th stimulus $\mathbf{s}^{(k)} = \begin{bmatrix} s_1^{(k)} & s_2^{(k)} & s_3^{(k)} & \dots & s_m^{(k)} \end{bmatrix}^T$ is paired with the k th target $\mathbf{t}^{(k)} = \begin{bmatrix} t_1^{(k)} & t_2^{(k)} & t_3^{(k)} & \dots & t_m^{(k)} \end{bmatrix}^T$ —the superscript (k) is simply a label indicating which pair we are referring to.

$$\begin{bmatrix} s_1^{(1)} & s_2^{(1)} & s_3^{(1)} & \dots & s_m^{(1)} \end{bmatrix}^T \rightarrow \begin{bmatrix} t_1^{(1)} & t_2^{(1)} & t_3^{(1)} & \dots & t_m^{(1)} \end{bmatrix}^T$$

$$\begin{bmatrix} s_1^{(2)} & s_2^{(2)} & s_3^{(2)} & \dots & s_m^{(2)} \end{bmatrix}^T \rightarrow \begin{bmatrix} t_1^{(2)} & t_2^{(2)} & t_3^{(2)} & \dots & t_m^{(2)} \end{bmatrix}^T$$

$$\begin{bmatrix} s_1^{(K)} & s_2^{(K)} & s_3^{(K)} & \dots & s_m^{(K)} \end{bmatrix}^T \rightarrow \begin{bmatrix} t_1^{(K)} & t_2^{(K)} & t_3^{(K)} & \dots & t_m^{(K)} \end{bmatrix}^T$$

A Matrix Memory for multiple pattern pairs encodes the weight matrix as the sum of the outer products of the target/stimulus pairs: $\mathbf{W} = \sum_{k=1}^K \mathbf{t}^{(k)} \otimes \mathbf{s}^{(k)} = \sum_{k=1}^K \mathbf{t}^{(k)} (\mathbf{s}^{(k)})^T$, where $(\mathbf{s}^{(k)})^T$ is the transpose of $\mathbf{s}^{(k)}$. Under this definition we have that the i, j th entry in $\mathbf{W}$ is given by:

$$W_{ij} = \sum_{k=1}^K t_i^{(k)} (s_j^{(k)})^T$$

The Matrix Memory takes a new stimulus vector $\mathbf{s}$ as input and computes the output response as $\mathbf{r} = \mathbf{W}\mathbf{s}$. If we wish to compute the response vector corresponding to the (l) th original pattern pair, we do the same $\mathbf{r}^{(l)} = \mathbf{W}\mathbf{s}^{(l)}$.

1. (5 points) **Prove that** for the l th response vector $\mathbf{r}^{(l)}$ for the l th stimulus $\mathbf{s}^{(l)}$ will equal the target $\mathbf{t}^{(l)}$ if all **pairs** of stimulus vectors $(\mathbf{s}^{(k)}, \mathbf{s}^{(l)}) \forall k \neq l$ are orthonormal to each other.

3 Eigenvectors and eigenvalues (7 points)

Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

with eigenvectors $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and corresponding eigenvalues $\lambda_1 = -1$ and $\lambda_2 = -2$.

1. (2 points) Compute the product Aw for $w = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, by directly performing the matrix multiplication.

2. (2 points) Now, express w in the basis given by the eigenvectors of A . That is, write w as a linear combination of the two eigenvectors.

3. (2 points) Using your answer to the question above, write the product Aw as a linear combination of the eigenvectors of A .

4. (1 point) Suppose that a matrix $B \in \mathbb{R}^{3 \times 3}$ has eigenvalues 4, -1, 0. Is B full rank, less than full rank, or is there not enough information to tell?

4 Positive semidefinite matrices (8 points)

During lecture, we learned that symmetric positive semidefinite (PSD) matrices do not “flip” vectors. In this problem, you will prove this claim using an eigenvector basis.

Let $A \in \mathbb{R}^{n \times n}$ be symmetric and PSD. You may use the fact that A has an orthonormal eigenbasis $v_1, \dots, v_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$. For a nonzero vector $x \in \mathbb{R}^n$, write

$$x = \sum_{i=1}^n a_i v_i.$$

1. (1 point) Express Ax in terms of a_i , λ_i , and v_i .

Hint: Apply A to the expression for x , using $Av_i = \lambda_i v_i$.

2. (2 points) Use part 1 to show that

$$x^T Ax = \sum_{i=1}^n \lambda_i a_i^2 \geq 0.$$

Hint: Expand the inner product. Orthonormality gives $v_i^T v_j = 0$ when $i \neq j$ and $v_i^T v_i = 1$.

3. (2 points) Suppose $Ax \neq 0$. The usual angle $\theta \in [0^\circ, 180^\circ]$ between x and Ax satisfies

$$\cos \theta = \frac{x^T Ax}{\|x\| \|Ax\|}.$$

Use part 2 to prove that $0^\circ \leq \theta \leq 90^\circ$. Explain how this establishes that multiplication by A does not “flip” x .

Hint: Determine the sign of the numerator and denominator.

4. (3 points) Show that $x^T Ax = 0$ if and only if $Ax = 0$. Can the angle in part 3 equal 90° ? Give an example of a nonzero symmetric PSD matrix A and a nonzero vector x with $Ax = 0$, and explain why the angle is undefined in this case.

Hint: A sum of nonnegative terms is zero only when every term is zero. For your example, consider a diagonal 2×2 matrix with a zero eigenvalue.

5 Collaboration Questions

After you have completed all other components of this assignment, report your answers to these questions regarding the collaboration policy. Details of the policy can be found in the syllabus.

1. Did you receive any help whatsoever from anyone in solving this assignment? If so, include full details.
2. Did you give any help whatsoever to anyone in solving this assignment? If so, include full details.
3. Did you find or come across code that implements any part of this assignment? If so, include full details.

Your Answer